

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		increases			1	1		
		(ii)		outer shell gets further away (from the nucleus) / there are more shells (1) there is lower attraction for an (incoming) electron / it is harder to gain an electron (1)	2			2		
	(b)			$\text{Cl}_2 + 2\text{NaBr} \rightarrow \text{Br}_2 + 2\text{NaCl}$ award (1) each for formulae Cl_2 NaCl award (1) for balancing only if both formulae are correct		3		3		
	(c)	(i)		3.20		1		1	1	
		(ii)		$\frac{1.27}{63.5} \quad \frac{3.20}{80} \quad (1)$ $0.02 : 0.04 \Rightarrow 1 : 2 \Rightarrow \text{CuBr}_2 (1)$ working must be shown ecf possible alternative method percentage composition of both elements		2		2	2	2